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### Internal combustion engine with ventilation system for a crankcase

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#### Abstract

Apparatuses, systems and methods are disclosed including internal combustion engine optionally including: a crankcase having a blow-by gas passing therethrough; a source of a boost air; a jacket containing a water; an oil separating apparatus in fluid communication with the boost air and the blow-by gas and having a coalescing filter to separate oil from the blow-by gas; and a heat exchanger in fluid communication with the boost air, wherein the heat exchanger is in fluid communication with the jacket to receive the water, wherein the water is passed in a heat transfer relationship with the boost air in the heat exchanger to achieve a desired temperature range for the boost air passing to the oil separating apparatus.

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**Inventors:** Smith; Lucas K. (Lafayette, IN), Lawrence; Rodney Allen (Frankfort, IN), Chambers; Joseph G. (Edwards, IL), Corcoran; Jarrett T. (Washington, IL), Stack; Brandyn A. (Lafayette, IN)

**Applicant:** Caterpillar Inc. (Peoria, IL)

**Family ID:** 95654630

**Assignee:** Caterpillar Inc. (Peoria, IL)

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*Primary Examiner:* Moubry; Grant

*Attorney, Agent or Firm:* Schwegman Lundberg & Woessner, P.A.

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## Background/Summary

### TECHNICAL FIELD

(1) The present disclosure relates to internal combustion engines such as those for vehicles or stationary power generation. More particularly, the present disclosure relates to heating and/or cooling boost air of a crankcase ventilation system for the internal combustion engine.

### BACKGROUND

(2) Machinery, for example, agricultural, industrial, construction or other heavy machinery can be propelled by an internal combustion engine(s). Internal combustion engines can be used for other purposes such as for power generation. Internal combustion engines combust a mixture of air and fuel in cylinders and thereby produce drive torque and power. Internal combustion engines typically include a crankcase to provide a housing for a crankshaft of the engine. A portion of the combustion gases (termed “blow-by”) may escape the combustion chamber past the piston and enter undesirable areas of the engine such as the crankcase. Crankcase ventilation systems exist. These systems can utilize filtration to reduce methane footprint, reduce particulate matter levels

and reduce loss of oil. Additionally, crankcase ventilation systems are known in internal combustion engines to vent blow-by gases within the crankcase. For example, U.S. Pat. No. 4,768,493, Japanese Patent Application Publication No. 2019178609 and United States Patent Application Publication No. 2023/0066495 disclose examples of crankcase ventilation systems. However, this patent and these patent applications do not provide for heating and/or cooling of engine boost air in the manner disclosed herein.

## SUMMARY

(3) In an example according to this disclosure, an internal combustion engine optionally including: a crankcase having a blow-by gas passing therethrough; a source of a boost air; a jacket containing a water; an oil separating apparatus in fluid communication with the boost air and the blow-by gas and having a coalescing filter to separate oil from the blow-by gas; and a heat exchanger in fluid communication with the boost air, wherein the heat exchanger is in fluid communication with the jacket to receive the water, wherein the water is passed in a heat transfer relationship with the boost air in the heat exchanger to achieve a desired temperature range for the boost air passing to the oil separating apparatus.

(4) In another example according to this disclosure, an engine system optionally including: a crankcase of an engine having a blow-by gas passing therethrough; a source of a boost air; a jacket containing a water; an oil separating apparatus in fluid communication with the boost air and the blow-by gas and having a coalescing filter to separate oil from the blow-by gas; and a heat exchanger in fluid communication with the boost air, wherein the heat exchanger is in fluid communication with the jacket to receive the water, wherein the water is passed in a heat transfer relationship with the boost air in the heat exchanger to achieve a desired temperature range for the boost air passing to the oil separating apparatus.

(5) In yet another example according to this disclosure, a method of maintaining a blow-by gas from a crankcase of an engine within a desired temperature range when passing through an oil separating apparatus is disclosed. The method optionally including: passing the blow-by gas from the crankcase to the oil separating apparatus; supplying a water from a jacket to a heat exchanger, wherein within the heat exchanger the water is in a heat transfer relationship with a boost air passing through the heat exchanger and heats or cools the boost air to a desired temperature range; passing the boost air at the desired temperature range to the oil separating apparatus; separating oil from the blow-by gas with the oil separating apparatus; passing the blow-by gas after leaving the oil separating apparatus to a charged air system; and returning the water to the jacket after passing through the heat exchanger.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

(1) In the drawings, which are not necessarily drawn to scale, like numerals may describe similar components in different views. Like numerals having different letter suffixes may represent different instances of similar components. The drawings illustrate generally, by way of example, but not by way of limitation, various embodiments discussed in the present document.

(2) FIG. 1 is a highly schematic illustration of an example internal combustion engine including a crankcase and a ventilation system for the crankcase including a heat exchanger for warming boost air in accordance with an example of this disclosure.

(3) FIG. 1A is a highly schematic illustration of an example internal combustion engine including air-to-air-aftercooled system as part of the ventilation system for the crankcase in accordance with an example of this disclosure.

(4) FIG. 2 is a perspective view of the ventilation system for the crankcase including the heat exchanger of FIG. 1.

(5) FIG. 3 is a highly schematic illustration of the internal combustion engine including the crankcase and another ventilation system for the crankcase including a second heat exchanger for warming the boost air and warming or cooling a blow-by gas in accordance with an example of this disclosure.

(6) FIG. 4 is a highly schematic illustration of the example internal combustion engine including the crankcase and yet another ventilation system for the crankcase configured for cooling boost air in accordance with an example of this disclosure.

#### DETAILED DESCRIPTION

(7) Examples according to this disclosure are directed to crankcase ventilation systems for supplying filtered blow-by gas to the internal combustion engine to separate oil from blow-by and re-ingest the blow-by into a charged air system. Examples of the present disclosure are now described with reference to the accompanying drawings. The following description is merely exemplary in nature and is not intended to limit the present disclosure, application, or use.

Examples described set forth specific components, devices, systems and methods, to provide an understanding of embodiments of the present disclosure. It will be apparent to those skilled in the art that specific details need not be employed and that examples may be embodied in many different forms. Thus, the examples provided should not be construed to limit the scope of the claims.

(8) As used herein, the terms “comprises,” “comprising,” “having,” “including,” or other variations thereof, are intended to cover a non-exclusive inclusion such that a process, method, article, or apparatus that comprises a list of elements does not include only those elements, but may include other elements not expressly listed or inherent to such a process, method, article, or apparatus. Further, relative terms, such as, for example, “about,” “substantially,” “generally,” and “approximately” are used to indicate a possible variation of  $\pm 15\%$  in a stated value.

(9) FIG. 1 depicts in a highly schematic manner of an internal combustion engine **100** (generally referred to as “engine” herein for simplicity) in accordance with this disclosure. The engine **100** can be used for power generation such as for the propulsion of vehicles or other machinery or for stationary power generation. The engine **100** can include various power generation platforms, and can use fuel including, for example, gasoline, gaseous fuel, diesel or blends thereof. Stationary engines may be used to drive immobile equipment, such as pumps, generators, mills, or factory equipment. In one embodiment, the engine **100** can be used in landfill applications for generating electricity. As such, the engine **100** may employ gaseous fuel. As used herein, “gaseous fuel” may include fuel that is supplied to the engine **100** in gaseous form and can include, for example, propane, natural gas, gas associated with natural gas such as bio-gas, landfill gas, carbon monoxide, hydrogen, hydrogen sulfide, or mixtures thereof. The fuel may have different levels of purity. As used herein, natural gas refers to both pure and relatively impure forms having various amounts of methane and other constituents. It is understood that the present disclosure can apply to any number of piston-cylinder arrangements and a variety of engine configurations including, but not limited to, V-engines, inline engines, and horizontally opposed engines, as well as overhead cam and cam-in-block configurations.

(10) According to some embodiments such as the embodiment of FIG. 1A, the engine **100A** can be air-to-air-aftercooled (“ATAAC”), (also called charge-air-cooled). As such, the engine **100A** uses forced-air (instead of water) to cool the turbocharged air before it enters the engine's combustion chamber. The engine **100A** can be constructed in a manner similar to the engine **100** of FIG. 1 including the various components discussed below in reference to FIG. 1 including a heat exchanger **112** and an oil separating apparatus **114**.

(11) The internal combustion engine **100** or **100A** can be used in stationary applications as discussed above but also can be used with vehicles and machinery that include those related to various industries, including, as examples, construction, agriculture, forestry, transportation, material handling, waste management, etc.

(12) Referring now again to FIG. 1, the engine **100** can be a dual turbo engine according to one example. The engine **100** can include a ventilation system **102** for a crankcase **104**, an aftercooler **106** and a jacket **108**. The ventilation system **102** can include one or more breathers **110**, a heat exchanger **112**, an oil separating apparatus **114**, a first turbocharger **116**, passages **118A**, **118B**, **118C**, **118D**, **118E** and **118F** and a second turbocharger **120**.

(13) A motive air for the ventilation system **102** can be at boost pressure. The motive air can be routed to the oil separating apparatus **114** (as further discussed and illustrated) or to a crankcase ventilation device and can then be routed to the first turbocharger **116**. The passages **118A**, **118B**, **118C**, **118D**, **118E** and **118F** allow for fluid communication between various components of the engine **100**. Thus, the passage **118A** fluidly connects the one or more breathers **110** with the oil separating apparatus **114**. The passage **118B** fluidly connects to the engine **100** downstream of or at the aftercooler **106** and fluidly communicates with the heat exchanger **112**. The passages **118C** and **118D** fluidly connect the jacket **108** with the heat exchanger **112**. The passage **118E** fluidly connects the heat exchanger **112** to the oil separating apparatus **114**. The passage **118F** fluidly connects the oil separating apparatus **114** with the first turbocharger **116**. Although not illustrated in FIG. 1, it is recognized that the ventilation system **102** can include valves or other regulators configured to regulate flow of fluid such as to prevent reverse flow when the pressure within the oil separating apparatus **114** and/or the heat exchanger **112** is lower than the pressure within the crankcase **104**. The terms “passage”, “passages”, “passageway”, “passageways”, “line” or “lines” as used herein should be interpreted broadly. These terms can be features defined by the various components of the engine illustrated in the FIGURES or can be direct interfacing connections or can be formed by additional components (e.g., a hose, tube, pipe, manifold, cavity etc.) as known in the art.

(14) In the example of FIG. 1, the one or more breathers **110** can be in fluid communication via the passage **118A** to communicate a blow-by gas containing oil mist to the oil separating apparatus **114**. Boost air from the engine **100** downstream or at the aftercooler **106** can be in fluid communication via the passage **118B** with the heat exchanger **112**. The heat exchanger **112** can be in fluid communication via the passage **118E** with the oil separating apparatus **114** to communicate the boost air from the heat exchanger **112** to the oil separating apparatus **114**. Thus, the heat exchanger **112** can be upstream of the oil separating apparatus **114** with respect to a flow direction of the boost air. The jacket **108** can be in fluid communication via the passages **118C** and **118D** with the heat exchanger to communicate a water with the heat exchanger **112**. The oil separating apparatus **114** can be in fluid communication via the passage **118F** with the first turbocharger **116** to communicate at least the filtered blow-by gas from the oil separating apparatus **114** to the first turbocharger **116**.

(15) Components of the ventilation system **102** such as the oil separating apparatus **114** and the heat exchanger **112** can be mounted to the engine **100**. However, it is contemplated that such components may be separate from the engine **100** (e.g., not mounted thereto other than via passages **118A**, **118B**, **118C**, **118D**, **118E** and **118F**). The ventilation system **102** or some components thereof can be part of the original manufacture of the engine **100** or can be a retrofitted system that is added to the engine **100** during maintenance, upgrade or the like. The ventilation system **102** can be in fluid communication with the crankcase **104** such as via the passages **118A** and additional passages or components not specifically shown. Although not illustrated, the ventilation system **102** can be configured to supply ambient air or air from a pressurized source to the crankcase **104**.

(16) The engine **100** can employ various components most not specifically illustrated. The engine **100** can include the aftercooler **106** for reducing the temperature of air utilized by systems of the engine **100** during operation. The aftercooler **106** can be downstream of and in fluid communication to receive relatively warmer air from the second turbocharger **120** (or other component such as a compressor), for example. The jacket **108** can be a source of liquid such as water used to cool various components of the engine **100** during operation. The water can

additionally be used during engine startup according to some examples. This water can typically be about 100 degrees Celsius, for example. A pump (not shown) or other flow generating device can be used to pass the water from the jacket **108** to the heat exchanger **112** along the passage **118C** and return the water to the jacket **108** from the heat exchanger **112** along the passage **118D**.

(17) Apparatuses such as the one or more breathers **110** can couple directly or indirectly to the engine block, and can be in fluid communication with the crankcase **104**. Each of the one or more breathers **110** can comprise a mechanism that separates some of the oil droplets and oil mist from the blow-by gas in order to prevent some the oil droplets and oil mist contained in the blow-by gas from being taken out of the crankcase **104** along the flow of the blow-by gas. By way of example, the one or more breathers **110** can include one or more separation mechanisms such as an oil separation valve, splasher plate, serpentine passage, mesh or other obstruction. The one or more breathers **110** can be an outlet allowing passage of fumes, blow-by constituents and/or aerosolized oil to the passage **118A** and/or atmosphere or another location such as away from the engine **100**.

(18) During engine startup and/or during extended engine idle, the boost air can be undesirably cold for general use by the ventilation system **102**. This is especially the case when the engine **100** is operating in a cold environment. In particular, undesirably cold boost air can be insufficient to maintain the blow-by gas above the dew point temperature. If the blow-by gas falls below the dew point temperature for the blow-by gas, there is a risk of condensation/freezing of the water vapor, forming emulsion or ice in one or more of the various components discussed above including the oil separating apparatus **114** and within the engine **100** itself. Condensation/freezing can lead to high pressure within the crankcase **104** and engine shutdown. As such, it is desirable that the boost air is warmed by the ventilation system **102** for use particularly in the oil separating apparatus **114** where the boost air is combined with the blow-by gas.

(19) The heat exchanger **112** can be a liquid-to-air heat exchanger such as a jacket heat exchanger, for example. The heat exchanger **112** can be configured to receive the boost air along passage **118B**. Additionally, the heat exchanger **112** can be configured to receive the water from the jacket **108** along passage **118C**. The heat exchanger **112** can be configured such that the water is passed in a heat transfer relationship with the boost air in the heat exchanger **112** to achieve a desired temperature range for the boost air passing to the oil separating apparatus **114**. The desired temperature range can be between about 70 degrees Celsius and about 90 degrees Celsius, for example.

(20) The ventilation system **102** can use the oil separating apparatus **114** or devices to filter oil (e.g., oil mist) from the blow-by gas to reduce volatile content in the blow-by gas. The oil separating apparatus **114** can contain a filter media such as a coalescing filter, desiccant, combinations thereof or other media. The blow-by gas can initially pass into a central passage or entry cavity of the oil separating apparatus **114**. The filter media is configured to separate a portion of the oil contained in the blow-by gas. The filter media can have a construction known in the art. As an example, the filter media can be constructed using a single or multi-layer synthetic coalescing filter media wound around a core, or pleated. As an example, the filter media can be a layered but highly compressed metal mesh that is used as a first stage in separating moisture and oil from the blow-by gas. As oil entrained blow-by gas encounters this randomly woven maze, large water and oil droplets coalesce. In operation, the oil separating apparatus **114** can utilize the pressurized boost air (generally at intake manifold pressure) to offset restriction through the oil separating apparatus **114**. Additionally, the boost air (heated or cooled to the desired temperature range within the heat exchanger **112**) can be utilized to warm or cool the blow-by gas within the oil separating apparatus **114** to maintain a desired temperature range for the blow-by gas. This desired temperature range for the blow-by gas can be above a dew point temperature of the blow-by gas and below a temperature at which one or more components of the oil separating apparatus **114** become inoperable.

(21) The first turbocharger **116** can be in fluid communication with the oil separating apparatus **114**

to receive the filtered blow-by gas and the boost air. The first turbocharger **116** can be used to draw or pull the blow-by gas through the oil separating apparatus **114**. The motive device **116** can pass the filtered blow-by gas and the boost air back to a charged air system (e.g., an aftercooler, manifold, combustion) to be routed again through the ventilation system **102**. The first turbocharger **116** can be device configured to compress air and pass the compressed air to an ATAAC (FIG. **1A**) or the aftercooler **106** (FIG. **1**), for example.

(22) FIG. **2** is a side view of a portion of the engine **100** including the ventilation system **102**, crankcase **104**, the aftercooler **106**, the jacket **108**, the heat exchanger **112**, the oil separating apparatus **114**, the first turbocharger **116** and the second turbocharger **120**. The one or more breathers **110** (FIG. **1**) and the passages **118A**, **118B**, **118C**, **118D**, **118E** and **118F** (FIG. **1**) are not illustrated in FIG. **2**. Rather the direction of flow of the boost air, the water and the blow-by gas are indicated with arrows in FIG. **2**.

(23) FIG. **3** schematically shows the engine **100** including the crankcase **104**, the aftercooler **106**, the jacket **108**, one or more breathers **110**, the oil separating apparatus **114**, the first turbocharger **116**, the passages **118A**, **118B**, **118C**, **118D**, **118E** and **118F**, and the second turbocharger **120** as previously discussed. The engine **100** includes a ventilation system **102A** that utilizes a heat exchanger **112A** design that is modified from that of the example of FIG. **1**. In particular, the heat exchanger **112A** is in fluid communication with both the boost air and the blow-by gas via passages **118A** and **118B**. The water from the jacket **108** is passed in a heat transfer relationship with both the boost air and the blow-by gas in the heat exchanger **112A**. This arrangement can raise (or lower) the temperatures of the boost air and the blow-by gas entering the oil separating apparatus **114** as desired. The ventilation system **102A** of FIG. **3** includes an additional passage **118G** for communicating the heated or cooled blow-by gas from the heat exchanger **112A** to the oil separating apparatus **114**.

(24) FIG. **4** schematically shows the engine **100** including the crankcase **104**, the aftercooler **106**, the jacket **108**, the one or more breathers **110**, the heat exchanger **112**, the oil separating apparatus **114**, the first turbocharger **116**, the passages **118B**, **118C**, **118D**, **118E** and **118F**, and the second turbocharger **120** as previously discussed. The engine **100** includes a ventilation system **102B** that is modified to have a passage **118H** that communicates relatively warmer boost air (as compared with the configurations of FIGS. **1-3**) to the heat exchanger **112**. In particular, the source of the boost air can be upstream of the aftercooler **106** such as at or downstream of the second turbocharger **120**, for example. The flow of the water from the jacket **108** can be reversed from the configuration of FIG. **1**. The water is passed in the heat transfer relationship with the boost air in the heat exchanger **112** to cool the boost air to achieve the desired temperature range for the boost air passing to the oil separating apparatus **114**. It is contemplated that the configuration of the system of FIG. **4** could also be used with the configuration of the system of FIG. **3** according to some examples.

## INDUSTRIAL APPLICABILITY

(25) In operation, the engine **100** can be configured to combust fuel to generate power. While typically efficient, a small portion of the blow-by gases may escape the combustion chamber past the piston and enter undesirable areas of the engine such as the crankcase **104**. The air the ventilation systems **102**, **102A** and **102B** uses acts to ventilate the crankcase **104** and other components. This ventilation can include use of the oil separating apparatus **114** to filter oil to remove the oil from the blow-by gas.

(26) Oil separating apparatuses containing coalescing filters are known, however, these have disadvantages. These devices typically lack cold climate capability. For example, in a cold climate water vapor can condense out and/or freeze within the oil separating apparatus or other components of the engine **100**. This condensed and/or frozen water vapor can restrict flow of blow-by gas, which can lead to unwanted pressure spikes within the engine.

(27) The present application recognizes a construction for the ventilation system(s) **102**, **102A**,

**102B** that utilizes the heat exchanger **112**, **112A** to cool, maintain, and/or warm boost air that is then passed to and utilized to warm or cool the blow-by gas within the oil separating apparatus **114**. This can allow the blow-by gas to achieve the desired temperature range including in the oil separating apparatus **114**. As an example, the boost air can be pre-heated in the heat exchanger **112** using the jacket water during engine startup and heated during engine operation (including idling) prior to entering the oil separating apparatus **114**. This improves operation of the oil separating apparatus **114** in cold climate as condensed water vapor, emulsions and/or frozen water vapor that restricts flow can be avoided. Thus, the design of the ventilation systems **102**, **102A** and **102B** can have improved temperature robustness. Thus, the present ventilation systems **102**, **102A** and **102B** can be configured to reduce or prevent water condensate, emulsion and/or freezing. Additionally, use of the water at a desired temperature (typically about 100 degrees Celsius) for heat exchange can eliminate the need for various control systems for the boost air, which can save cost and reduce system complexity. The temperature of the water is desirable as it will not raise the boost air to an undesirably high temperature that could lead to temperature degradation amongst the polymer and elastomer components of the oil separating apparatus **114**, for example.

(28) The above detailed description is intended to be illustrative, and not restrictive. The scope of the disclosure should, therefore, be determined with references to the appended claims, along with the full scope of equivalents to which such claims are entitled.

## Claims

1. An internal combustion engine comprising: a crankcase having a blow-by gas passing therethrough; a source of a boost air; a jacket containing a water; an oil separating apparatus in fluid communication with the boost air and the blow-by gas and having a coalescing filter to separate oil from the blow-by gas; and a heat exchanger in fluid communication with the boost air, wherein the heat exchanger is in fluid communication with the jacket to receive the water, wherein the water is passed in a heat transfer relationship with the boost air in the heat exchanger to achieve a desired temperature range for the boost air passing to the oil separating apparatus.
2. The engine of claim 1, wherein the desired temperature range for the boost air to the oil separating apparatus is between about 70 degrees Celsius and 90 degrees Celsius.
3. The engine of claim 1, wherein the desired temperature range of the boost air within the oil separating apparatus is sufficient to maintain the blow-by gas above a dew point temperature of the blow-by gas and below a temperature at which one or more components of the oil separating apparatus become inoperable.
4. The engine of claim 1, wherein the source of the boost air is downstream of an aftercooler of the internal combustion engine.
5. The engine of claim 1, wherein the water is passed in the heat transfer relationship with the boost air in the heat exchanger to one of: warm or cool the boost air to achieve the desired temperature range for the boost air passing to the oil separating apparatus.
6. The engine of claim 1, wherein the heat exchanger is in fluid communication with the blow-by gas, and wherein the water is passed in a heat transfer relationship with the blow-by gas in the heat exchanger.
7. The engine of claim 1, wherein the source of the boost air is upstream of an aftercooler of the internal combustion engine.
8. The engine of claim 7, wherein the source of the boost air is a compressor of the internal combustion engine.
9. An engine system comprising: a crankcase of an engine having a blow-by gas passing therethrough; a source of a boost air; an aftercooler; a jacket containing a water; an oil separating apparatus in fluid communication with the boost air and the blow-by gas and having a coalescing filter to separate oil from the blow-by gas; and a heat exchanger in fluid communication with the



boost air, wherein the heat exchanger is in fluid communication with the jacket to receive the water, wherein the water is passed in a heat transfer relationship with the boost air in the heat exchanger to achieve a desired temperature range for the boost air passing to the oil separating apparatus; and either the source of the boost air is downstream of the aftercooler, and the water is passed in the heat transfer relationship with the boost air in the heat exchanger to warm the boost air to achieve the desired temperature range for the boost air passing to the oil separating apparatus or the source of the boost air is upstream of the aftercooler, and the water is passed in the heat transfer relationship with the boost air in the heat exchanger to cool the boost air to achieve the desired temperature range for the boost air passing to the oil separating apparatus.

10. The engine system of claim 9, wherein the heat exchanger is in fluid communication with the blow-by gas, and wherein the water is passed in a heat transfer relationship with the blow-by gas in the heat exchanger.

11. The engine system of claim 9, wherein the desired temperature range of the boost air within the oil separating apparatus is sufficient to maintain the blow-by gas above a dew point temperature of the blow-by gas and below a temperature at which one or more components of the oil separating apparatus become inoperable.

12. A method of maintaining a blow-by gas from a crankcase of an engine within a desired temperature range when passing through an oil separating apparatus, the method comprising: passing the blow-by gas from the crankcase to the oil separating apparatus; supplying a water from a jacket to a heat exchanger, wherein within the heat exchanger the water is in a heat transfer relationship with a boost air passing through the heat exchanger and heats or cools the boost air to a desired temperature range; passing the boost air at the desired temperature range to the oil separating apparatus; separating oil from the blow-by gas with the oil separating apparatus; passing the blow-by gas after leaving the oil separating apparatus to a charged air system; and returning the water to the jacket after passing through the heat exchanger.

13. The method of claim 12, wherein the boost air is supplied in an amount that offsets a flow restriction of the blow-by gas through the oil separating apparatus.

14. The method of claim 12, wherein the desired temperature range for the blow-by gas is above a dew point temperature of the blow-by gas and below a temperature at which one or more components of the oil separating apparatus become inoperable.

15. The method of claim 12, further comprising supplying the boost air from one of upstream or downstream of an aftercooler of the engine.

16. The method of claim 12, further comprising passing the water in a heat transfer relationship with the blow-by gas in the heat exchanger.

17. The method of claim 12, wherein the desired temperature range for the boost air to the oil separating apparatus is between about 70 degrees Celsius and 90 degrees Celsius.

18. The method of claim 12, further comprising supplying the boost air from a compressor of the engine.

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